

COMP 102 – Sets, Operations, Power Sets, Partitions & Relations (Weeks 4–5)

Kisii University | Clean Revision Notes

1. Proper Subsets

Definition: A set **X** is a **proper subset** of set **Y** if every element of X is also an element of Y, **and** X is not equal to Y. Written $X \subset Y$ (strict inclusion). Equivalently: $X \subseteq Y$ and $|X| < |Y|$.

Example: Let $X = \{1, 2, 3, 4, 5, 6\}$ and $Y = \{1, 2, 3\}$. Then $Y \subset X$ because every element of Y is in X and X has at least one extra element.

Note: The empty set $\emptyset$ is a proper subset of every non-empty set.

2. Singleton (Unit) Set

A set that contains **exactly one element**. Denoted $\{a\}$. Example: $S = \{x \in \mathbb{N} \mid 7 < x < 9\} = \{8\}$.

3. Equal Sets & Equivalent Sets

Equal sets: Two sets contain exactly the same elements (order and repetition do not matter). $A = B \Leftrightarrow (\forall x)(x \in A \leftrightarrow x \in B)$.

Example: $A = \{1, 2, 6\}$, $B = \{6, 1, 2\} \Rightarrow A = B$ (and $|A| = |B|$).

Equivalent sets: Have the same cardinality ($|A| = |B|$) but may contain different elements. Example: $A = \{1, 2, 6\}$, $B = \{13, 16, 10\} \Rightarrow |A| = |B| = 3$, so they are equivalent but not equal.

4. Universal Set

The collection of **all elements under consideration** in a particular context. Denoted **U** (or ξ). Every set discussed in that context is a subset of U.

Example: If U = set of all animals on Earth, then the set of all mammals $\subseteq U$ and the set of all insects $\subseteq U$.

5. Empty (Null) Set

A set that contains **no elements**. Denoted $\emptyset$ or $\{\}$. Its cardinality is zero: $|\emptyset| = 0$. It is a subset of every set.

6. Overlapping Sets

Two sets that have **at least one element in common**. Their intersection is non-empty: $A \cap B \neq \emptyset$.

Cardinality formulae (very useful for exams):

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- $n(A \cup B) = n(A - B) + n(B - A) + n(A \cap B)$
- $n(A) = n(A - B) + n(A \cap B)$
- $n(B) = n(B - A) + n(A \cap B)$

Example: $A = \{1, 2, 6\}$, $B = \{6, 12, 42\}$. Common element 6 $\Rightarrow$ overlapping sets.

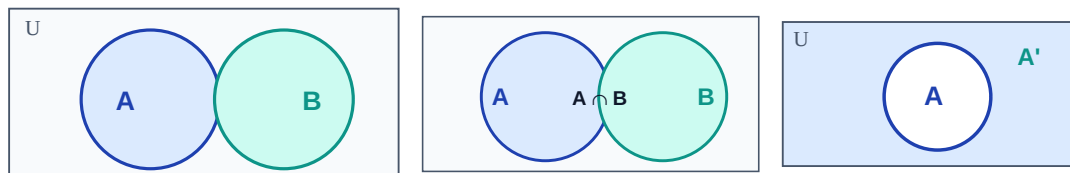
7. Disjoint Sets

Two sets A and B are **disjoint** if they have **no element in common**: $A \cap B = \emptyset$. Then $n(A \cup B) = n(A) + n(B)$.

Example: $A = \{1, 2, 6\}$, $B = \{7, 9, 14\}$. No common element $\Rightarrow$ disjoint.

8. Venn Diagrams

Schematic diagrams invented by John Venn that show all possible logical relations between sets. Extremely useful for visualising union, intersection, difference and complement.



Two sets (overlap) | Intersection region | Complement of A in U

9. Set Operations

9.1 Union ($A \cup B$)

The set of all elements that belong to A, to B, or to both. $A \cup B = \{x \mid x \in A \vee x \in B\}$.

Example: $A = \{10, 11, 12, 13\}$, $B = \{13, 14, 15\} \Rightarrow A \cup B = \{10, 11, 12, 13, 14, 15\}$.

9.2 Intersection ($A \cap B$)

The set of all elements that belong to **both** A and B. $A \cap B = \{x \mid x \in A \wedge x \in B\}$.

Example: $A = \{11, 12, 13\}$, $B = \{13, 14, 15\} \Rightarrow A \cap B = \{13\}$.

9.3 Complement of a Set (A' or A^c)

Relative to a universal set U , the complement of A is the set of all elements of U that are **not** in A . $A' = \{x \in U \mid x \notin A\}$.

Example: If $U = \{10, 11, 12, 13, 14, 15\}$ and $A = \{11, 12, 13\}$, then $A' = \{10, 14, 15\}$.

9.4 Set Difference / Relative Complement ($A - B$ or $A \setminus B$)

Elements that are in A but **not** in B . $A - B = \{x \mid x \in A \wedge x \notin B\}$. Note carefully: $A - B \neq B - A$ in general.

Example: $A = \{10, 11, 12, 13\}$, $B = \{13, 14, 15\} \Rightarrow A - B = \{10, 11, 12\}$, $B - A = \{14, 15\}$.

9.5 Cartesian Product (Cross Product)

The set of all ordered pairs (x, y) where $x \in A$ and $y \in B$. Written $A \times B$. For n sets: $A_1 \times A_2 \times \dots \times A_n$ is the set of all ordered n -tuples.

Example: $A = \{a, b\}$, $B = \{1, 2\} \Rightarrow A \times B = \{(a,1), (a,2), (b,1), (b,2)\}$. Note $B \times A = \{(1,a), (1,b), (2,a), (2,b)\}$ which is different. We often write A^2 for $A \times A$.

10. Power Set

The power set of S , denoted $\mathbf{P(S)}$ or 2^S , is the set of **all subsets** of S , including $\emptyset$ and S itself. If $|S| = n$ then $|P(S)| = 2^n$.

Example: $S = \{a, b, c, d\}$, $|S| = 4 \Rightarrow |P(S)| = 2^4 = 16$. The subsets include $\emptyset$, $\{a\}$, $\{b\}$, ..., $\{a,b\}$, ..., $\{a,b,c,d\}$.

11. Partitioning of a Set & Bell Numbers

A **partition** of a set S is a collection of non-empty, pairwise disjoint subsets $P_1, P_2, \dots, P_k$ whose union is exactly S . That is:

- $P_i \neq \emptyset$ for every i
- $P_i \cap P_j = \emptyset$ whenever $i \neq j$
- $P_1 \cup P_2 \cup \dots \cup P_k = S$

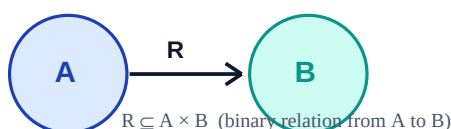
Bell number B_n counts the number of different partitions of a set with n elements. Important values: $B_0 = 1$, $B_1 = 1$, $B_2 = 2$, $B_3 = 5$, $B_4 = 15$, $B_5 = 52$.

All 5 partitions of $S = \{1, 2, 3\}$ ($B_3 = 5$):

1. $\{\{1\}, \{2\}, \{3\}\}$ (three singletons)
2. $\{\{1, 2\}, \{3\}\}$
3. $\{\{1, 3\}, \{2\}\}$
4. $\{\{1\}, \{2, 3\}\}$
5. $\{\{1, 2, 3\}\}$ (one block – the whole set)

12. Relations (Binary Relations)

In mathematics we often consider the existence or non-existence of a connection between ordered pairs of objects. Formally, a **binary relation R** from set A to set B is a **subset of the Cartesian product $A \times B$** . We write $a R b$ (or $(a, b) \in R$) when a is related to b .



$R \subseteq A \times B$ (binary relation from A to B)

If $R \subseteq A \times A$ we say R is a relation on A . The empty relation and the universal relation $A \times A$ are always possible.

12.1 Domain and Range

- **Domain of R** = $\{a \in A \mid \exists b \in B \text{ such that } (a, b) \in R\}$ (the first components that actually appear).
- **Range of R** = $\{b \in B \mid \exists a \in A \text{ such that } (a, b) \in R\}$ (the second components that appear).

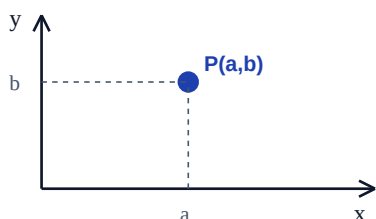
Example: $A = \{1, 2, 3\}$, $B = \{x, y, z\}$, $R = \{(1, x), (1, y), (2, y), (3, z)\}$. Then $\text{domain}(R) = \{1, 2, 3\}$, $\text{range}(R) = \{x, y, z\}$.

12.2 Set Inclusion as a Relation

For any collection of sets, the relation $\subseteq$ is a binary relation: $A \subseteq B$ means (A, B) belongs to the inclusion relation. It is reflexive, antisymmetric and transitive (a partial order).

12.3 Geometric View of R^2 (Cartesian Plane)

R denotes the set of real numbers; $R^2 = R \times R$ is the set of all ordered pairs of real numbers. Each point P in the plane represents an ordered pair (a, b) . The vertical line through P meets the x -axis at a ; the horizontal line meets the y -axis at b . R^2 is frequently called the Cartesian plane.



Point $P(a, b)$ in the Cartesian plane R^2

Quick Revision Checklist

- Proper subset: $X \subset Y$ means $X \subseteq Y$ and $X \neq Y$ (strictly smaller).
- Singleton = set with exactly one element; Empty set $\emptyset$ has zero elements.
- Equal sets have identical elements; Equivalent sets have same cardinality only.
- Universal set U contains every element under discussion; all other sets $\subseteq U$.
- Overlapping $\Leftrightarrow A \cap B \neq \emptyset$; Disjoint $\Leftrightarrow A \cap B = \emptyset$.
- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ (inclusion-exclusion).
- $A - B = \{x \in A \mid x \notin B\}$; remember $A - B \neq B - A$ in general.
- Complement $A' = U - A$.
- $|P(S)| = 2^n$; power set contains every possible subset.
- Partition: non-empty, pairwise disjoint subsets whose union is the whole set.
- Bell number B_n = number of partitions of an n -element set ($B_3 = 5$).
- Binary relation R from A to B is any subset of $A \times B$.
- Domain = first components used; Range = second components used.
- Cartesian product $A \times B$ = all ordered pairs (a, b) ; order matters.